

TURNING POINT

MACHINE-GUNS

In 1883, a patent filed by Hiram Maxim covered a revolutionary concept in firearms—a gun in which energy from the explosion, in addition to driving the bullet, also drove the cycle of loading and firing and would continue to do so as long as there was a supply of ammunition. All the user had to do was point the gun and pull the trigger. This true machine-gun inspired the development of new “fully automatic” and “semiautomatic” firearms, which are the weapons used today by the world’s armed forces and law enforcement agencies.



▲ MAXIM GUN

In the Maxim gun, the energy from the recoil was used to eject each spent cartridge and insert the next one and fire it. This made it less labor-intensive and more efficient than previous rapid-firing guns that relied on manual cranking.

Mechanized guns, or what were considered to be “machine-guns,” first began to appear in the middle of the 19th century. Their operation involved feeding the cartridge into the chamber of a barrel, firing it, and then extracting the empty case by a manually powered mechanical process in a continuous cycle. The first of these guns to achieve real success was the Gatling, later followed by the Nordenfelt, Hotchkiss, and Gardner guns. All performed well in their

own distinctive ways and were widely used by the major military and maritime nations of the world. However, all mechanical machine-guns suffered from the same drawback—they required human energy to operate them and stamina to maintain a continuous fire.

RECYCLING WASTED ENERGY

All guns obey the same law of nature—the force that drives the ammunition forward also drives the gun backward. This was considered an unavoidable nuisance by gunmakers. Hiram Maxim, however, recognized it as a source of energy and put it to better use. He also

noted other flaws associated with machine-guns, which included cartridges that often suffered from a “hang-fire”—a delayed explosion of the main charge after the primer was detonated.

THEORY BECOMES REALITY

Maxim experimentally modified rifles to use their recoil energy to load and fire them. Satisfied that the idea could work, he built an experimental gun, which operated in the same way but used a specially designed lock mechanism. This mechanism extracted cartridges from a continuous belt, fed them into the chamber, and fired them. A hang-fire was not a problem in such a gun, because it could not continue its cycle until the explosion occurred.

Conscious of the heat generated by continuous

» BEFORE

The third quarter of the 19th century saw the creation of guns capable of giving sustained fire. Often referred to at the time as “battery guns,” they became thought of as “machine-guns” because the processes of loading and firing had been mechanized, turning them into “shooting machines.” They were successful, but they had their drawbacks.



GATLING GUN

- **MOST GUNS WERE HEAVY** and often needed to be mounted on wheeled carriages for transport. Their use on land and at sea needed massive mountings fixed to the deck or other structures capable of supporting them.
- **A CREW OF SEVERAL MEN** was required to operate the guns, and a team of horses had to draw the gun carriage along with a limber to carry the ammunition.
- **LIGHTWEIGHT, PORTABLE MACHINE-GUNS**, such as the Nordenfelt gun, were developed. However, since they were hand-cranked, their aim was easily disturbed and not very accurate.

